Indian Institute of Information Technology Design and Manufacturing, Kancheepuram
Department of Electronics and Communication Engineering

Subject: Basic Electrical Engineering
Instructor: Dr. B. Chitti Babu
Date: 25/09/2025
Due Date: 06/10/2025

Assignment I

Basic concepts

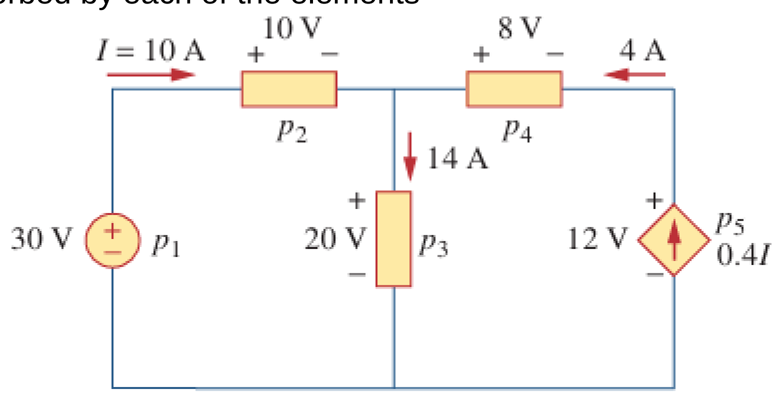
1. A 60-W incandescent lamp is connected to a 120-V source and is left burning continuously in an otherwise dark staircase. Determine:
(a) The current through the lamp.
(b) the cost of operating the light for one non-leap year if electricity costs 9.5 paise per kWh.

2. A flashlight battery has a rating of 0.8 ampere-hours (Ah) and a lifetime of 10 hours.
(a) How much current can it deliver?
(b) How much power can it give if its terminal voltage is 6 V?
(c) How much energy is stored in the battery in Wh?

3. (a) A lightning bolt with 10 kA strikes an object for 15 µs. How much charge is deposited on the object?
(b) A rechargeable flashlight battery is capable of delivering 90 mA for about 12 h. How much charge can it release at that rate? If its terminal voltage is 1.5 V, how much energy can the battery deliver?

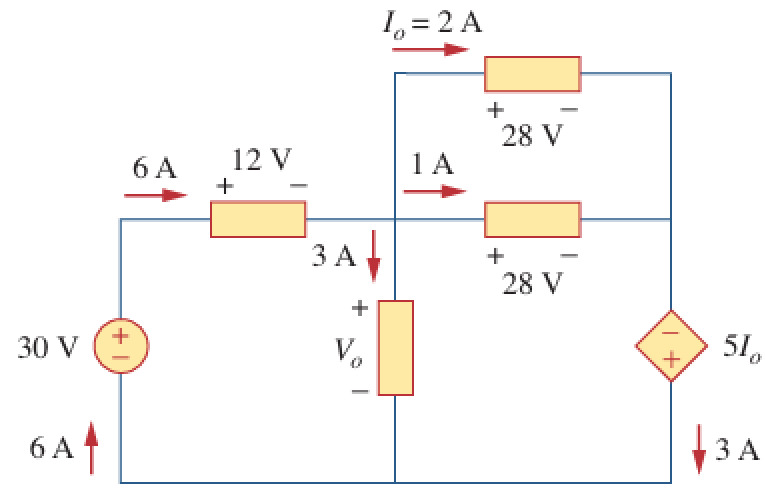
4. The charge entering the positive terminal of an element is $q = 5\sin 4\pi t \; mC$ while the voltage across the element (plus to minus) is $v = 3\cos 4\pi t \; V$
(a) Find the power delivered to the element at t = 0.3 s.
(b) Calculate the energy delivered to the element between 0 and 0.6 s.

5. Find the power absorbed by each of the elements

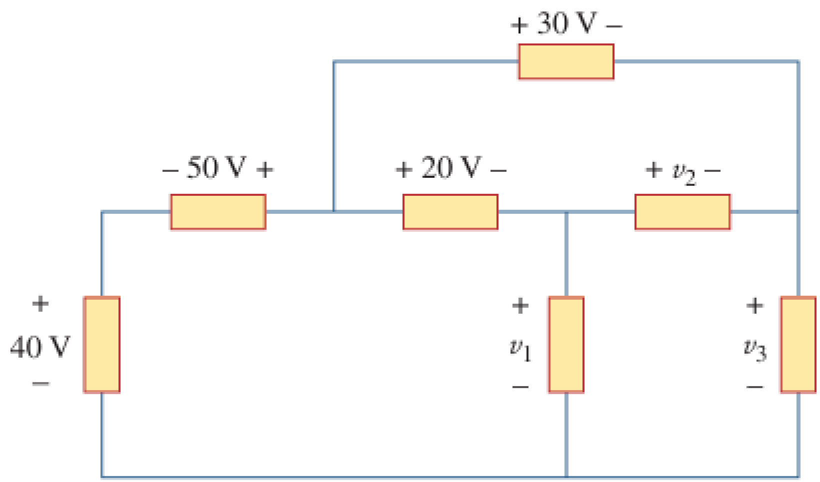


Basic Laws

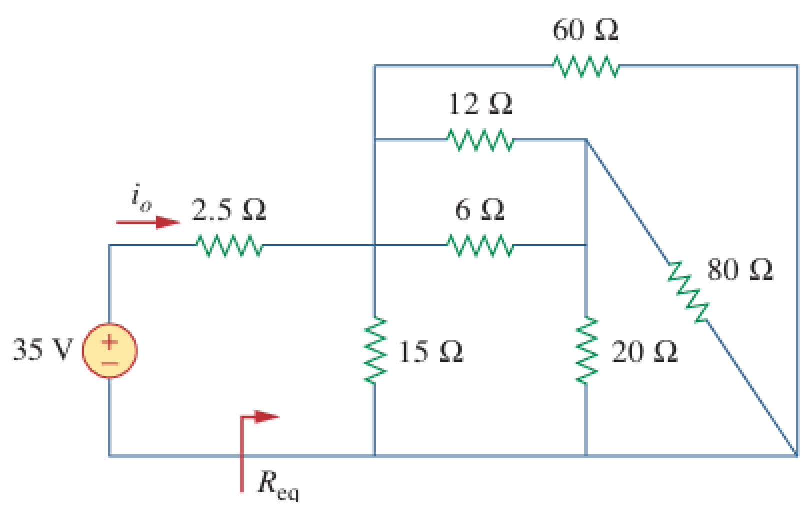
6. Find V_o and the power absorbed by each element in the circuit



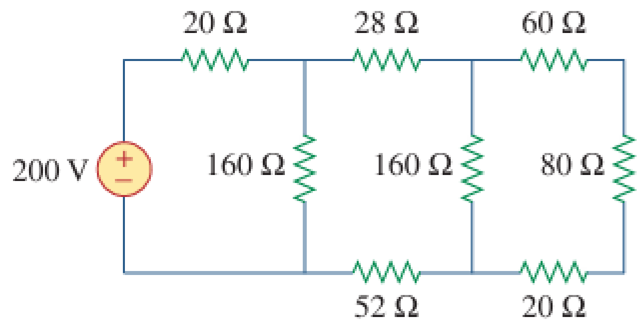
7. In the circuit, obtain v_1, v_2 and v_3.



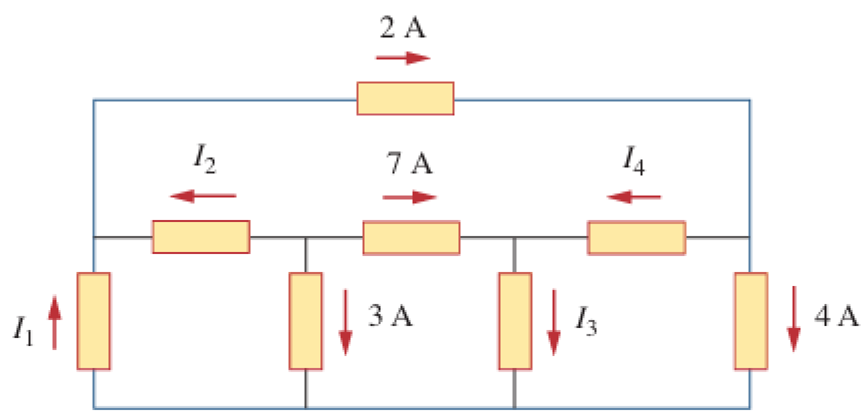
8. Find R_{eq} and i_o in the circuit



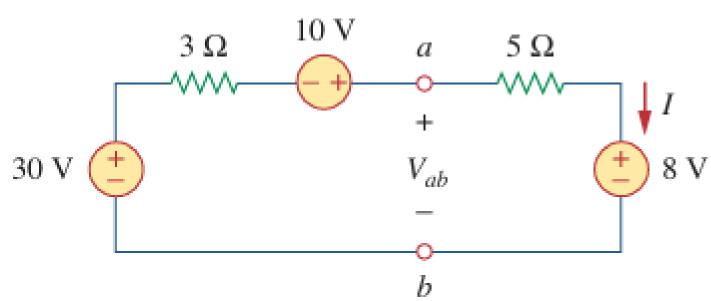
9. Using series/parallel resistance combination, find the equivalent resistance seen by the source in the circuit of Figure. Find the overall absorbed power by the resistor network.



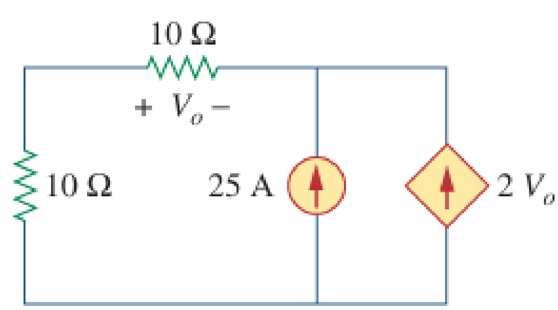
10. For the circuit in Figure, use KCL to find the branch currents I_1 to I_4.



11. Find I and V_{ab} in the circuit

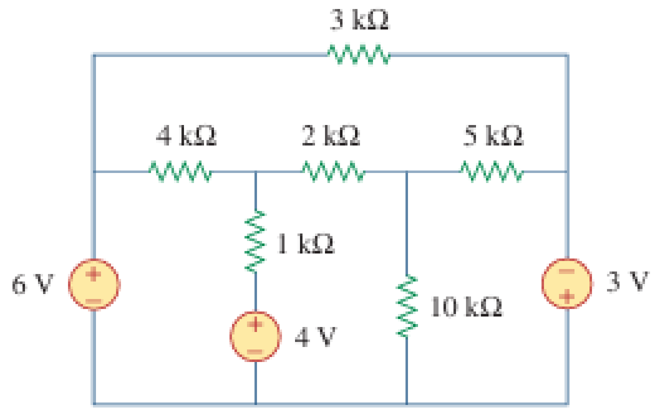


12. Find Vo in the circuit in Figure and the power absorbed by the dependent source.

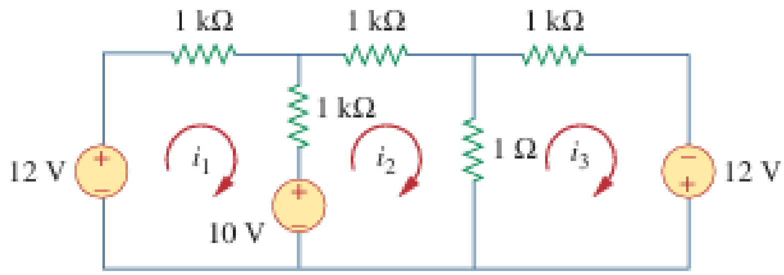


MESH ANALYSIS

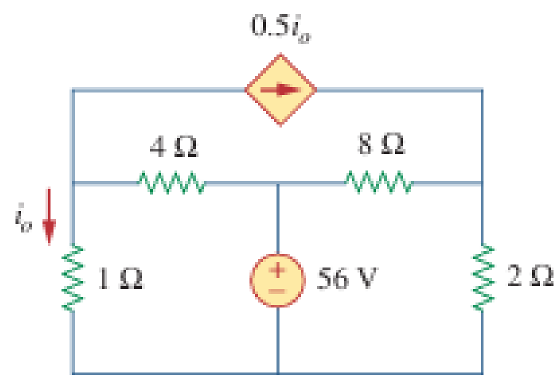
13. Determine the current through the 10-k resistor in the circuit using mesh analysis



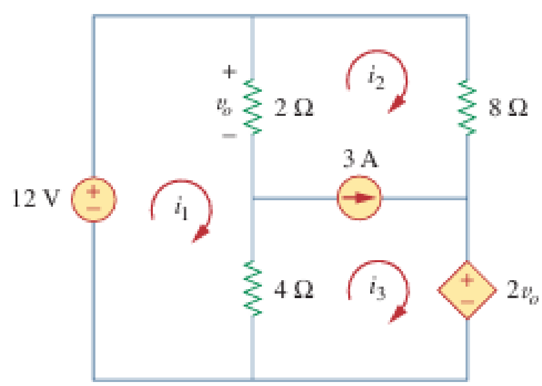
14. Find the mesh currents i_1, i_2, and i_3 in the circuit



15. Calculate the power dissipated in each resistor in the circuit

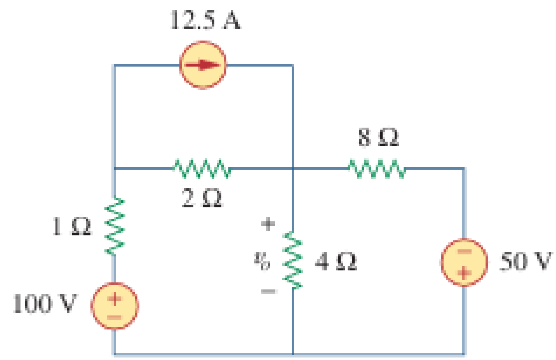


16. Use mesh analysis to find i_1, i_2, and i_3 in the circuit

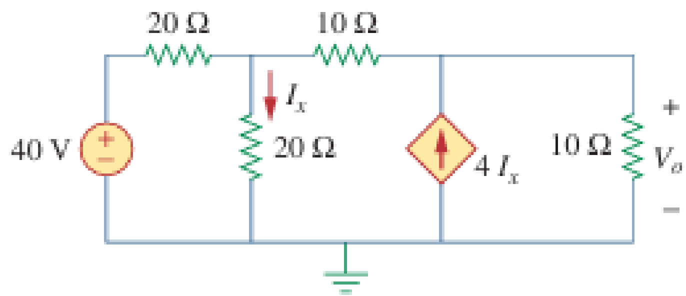


NODAL ANALYSIS

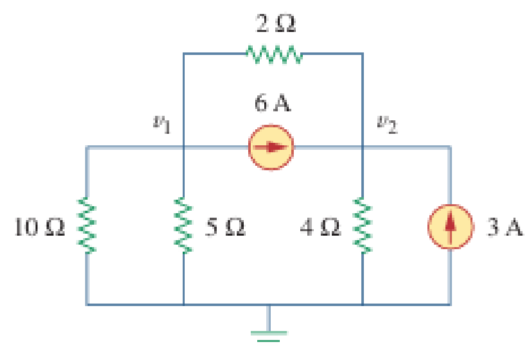
17. Using nodal analysis, find v_0 in the circuit



18. Using nodal analysis, find v_0 in the circuit of



19. For the circuit, Obtain V_1 and V_2



20. Determine the node voltages in the circuit